

SVG Tutorial

[◀ Previous](#)[Next ▶](#)

[REMOVE ADS](#)

SVG stands for Scalable Vector Graphics.

SVG defines vector-based graphics in XML format.

SVG graphics are scalable, and do not lose any quality if they are zoomed or resized.

SVG is supported by all major browsers.

Examples in Each Chapter

With our "Try it Yourself" editor, you can edit the SVG, and click on a button to view the result.

SVG Example

```
<html>
<body>

<h1>My first SVG</h1>

<svg width="100" height="100" xmlns="http://
www.w3.org/2000/svg">
  <circle cx="50" cy="50" r="40" stroke="green"
stroke-width="4" fill="yellow" />
```

[REMOVE ADS](#)
[Try it Yourself »](#)

What you should already know

Before you continue, you should have a basic understanding of the following:

- [HTML](#)
- [Basic XML](#)

If you want to study these subjects first, find the tutorials on our [Home page](#).

What is SVG?

- SVG stands for Scalable Vector Graphics
- SVG is used to define vector-based graphics for the Web
- SVG defines graphics in XML format
- Each element and attribute in SVG files can be animated
- SVG is a W3C recommendation
- SVG integrates with other standards, such as CSS, DOM, XSL and JavaScript

[REMOVE ADS](#)[REMOVE ADS](#)

SVG is a W3C Recommendation

SVG 1.0 became a W3C Recommendation on 4 September 2001.

SVG 1.1 became a W3C Recommendation on 14 January 2003.

SVG 1.1 (Second Edition) became a W3C Recommendation on 16 August 2011.

SVG Advantages

Advantages of using SVG over other image formats (like JPEG and GIF) are:

- SVG images are scalable
- SVG images can be printed with high quality at any resolution
- SVG images are zoomable
- SVG graphics do NOT lose any quality if they are zoomed or resized
- SVG is an open standard
- SVG files are pure XML

How does it Work?

SVG has elements and attributes for rectangles, circles, ellipses, lines, polygons, curves, and more.

SVG also supports filter and blur effects, gradients, rotations, animations, interactivity with JavaScript, and more.

A simple SVG document consists of the `<svg>` root element and several basic shape elements that will build a graphic together.

Creating SVG Images

SVG images can be created with any text editor, or with a drawing program, like [Inkscape](#).

For you to learn the concept and basics of SVG, this tutorial will just use plain text to teach you SVG.

[REMOVE ADS](#)[◀ Previous](#)[Sign in to track progress](#)[Next ▶](#)

Get W3Schools in Your Classroom

[Learn more](#)

Top Tutorials

[HTML Tutorial](#)[CSS Tutorial](#)

[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HTML](#)[PHP Tutorial](#)[Java Tutorial](#)[FORUM](#)[ABOUT](#)[ACADEMY](#)[C++ Tutorial](#)[jQuery Tutorial](#)

W3Schools is optimized for **SPACES** and training. Examples might be slow for reading and learning.

Top References

Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness

of all content. While using W3Schools, you agree to have read and accepted our [terms of use](#), [cookies](#) and [privacy policy](#).

[HTML Reference](#)[CSS Reference](#)[JavaScript Reference](#)[SQL Reference](#)

Copyright 1999-2025 W3Schools Data. All Rights Reserved. [W3Schools is](#)

[Python Reference](#)[W3.CSS Reference](#) Powered by [W3.CSS](#).[Bootstrap Reference](#)[PHP Reference](#)[HTML Colors](#)[Java Reference](#)

FOR TEACHERS [AngularJS Reference](#)

[jQuery Reference](#)

Top Examples

[HTML Examples](#)[CSS Examples](#)

BOOTCAMP'S [JavaScript Examples](#)

[How To Examples](#)[SQL Examples](#)[Python Examples](#)[W3.CSS Examples](#)[Bootstrap Examples](#)[PHP Examples](#)

CONTACT'S [Java Examples](#)

[XML Examples](#)[jQuery Examples](#)

Get Certified

[HTML Certificate](#)[CSS Certificate](#)[JavaScript Certificate](#)[Front End Certificate](#)[SQL Certificate](#)[Python Certificate](#)[PHP Certificate](#)[jQuery Certificate](#)[Java Certificate](#)

[REMOVE ADS](#)